

NORTHPORT 2 S, AL, USA

WMO: 722288

Lat: 33.2125N

Lon: 87.5914W

Elev: 46

StdP: 100.78

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 73801

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.6	(c) -3.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.8	(c) 35.9	(d) N/A	(e) 34.7	(f) N/A	(g) 33.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-9.1	37.2	2.8	1.8	-11.1	38.6	-12.7	39.6	-14.3	40.7	-16.3	42.0
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	7.1	10.1	13.7	18.0	22.3	26.5	27.5	27.2	24.9	18.6	12.2	9.1
	DBStd	8.36	5.94	5.76	5.14	3.63	3.59	2.11	1.79	1.97	3.08	4.52	4.77	5.05
	HDD10.0	332	128	66	24	1	0	0	0	0	1	31	82	
	HDD18.3	1344	351	238	158	50	12	0	0	1	53	192	289	
	CDD10.0	3300	38	67	139	239	382	496	542	534	447	267	96	53
	CDD18.3	1269	1	6	14	39	136	246	284	276	198	61	7	2
	CDH23.3	12287	3	24	107	367	1243	2517	2871	2673	1852	588	37	5
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	1337	135	129	144	124	102	108	123	92	82	81	112	120
	PrecMax	1915	245	368	450	381	283	304	312	189	222	231	289	263
	PrecMin	900	28	33	30	10	11	9	22	19	20	3	25	25
	PrecStd	244	51	70	81	73	66	64	67	45	57	58	57	54
	DB	23.6	26.1	28.4	29.8	33.8	37.4	36.9	37.3	36.1	33.0	26.9	24.0	
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	21.0	23.7	26.2	28.4	32.2	35.3	35.2	35.8	34.6	30.7	24.0	21.5	
	2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% DB	18.8	21.4	24.2	27.1	30.8	33.9	34.0	34.2	33.2	28.9	22.3	19.5	
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10% DB	16.7	19.4	22.1	25.6	29.6	32.7	32.8	32.8	31.7	26.8	20.5	17.5	
	10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(35) Mean Daily Temperature Range	MDBR	11.3	11.2	12.0	12.6	12.0	11.5	10.8	10.8	12.0	13.0	13.0	10.3	
	5% DB	12.2	12.1	13.3	13.6	13.3	13.3	12.2	13.0	13.6	14.9	13.5	11.8	
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(40) Clear-Sky Solar Irradiance	taub	0.324	0.335	0.365	0.402	0.451	0.482	0.534	0.533	0.443	0.374	0.347	0.324	
	taud	2.519	2.482	2.418	2.303	2.212	2.155	2.033	2.029	2.269	2.452	2.465	2.534	
	Ebn at Noon	893	918	912	888	843	813	769	762	822	863	856	873	
	Edh at Noon	84	97	111	130	144	152	171	169	128	99	88	79	
(44) All-Sky Solar Radiation	RadAvg	2.65	3.18	4.34	5.50	6.06	6.28	6.04	5.55	4.91	4.07	3.01	2.27	
	RadStd	0.21	0.44	0.40	0.43	0.45	0.49	0.38	0.34	0.42	0.50	0.31	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (35 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+143	+103

Nomenclature: See separate page